



Project report

Analysis of “Netflix Customer
Churn & Engagement
Analytics” Dataset

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Content

- Data Preparation
- Descriptive Statistics
- Hypothesis Testing
- Predictive Modeling
- Decision Trees
- K-Means & K-Medoids



Dataset Description

Selected Dataset



ZEYAD MOHAMED · UPDATED A MONTH AGO

▲ 48 <> Code ⬇ Download ⋮

Netflix Customer Churn & Engagement Analytics

5,000 Netflix subscribers dataset for churn prediction, retention analytics, and

The Netflix logo, a large red 'N' on a black background.

A comprehensive, curated dataset analyzing customer churn dynamics, behavioral engagement, subscription economics, and demographics for **5,000 Netflix subscribers** across 6 global continents.

< Link : <https://www.kaggle.com/datasets/zeyadmohamed26/netflix-customer-churn-and-engagement-analytics/data> >



Attributes Description *(Part 1)*

customer_id	string	UUID format (e.g. a9b75100-...)	Unique anonymous subscriber identifier.
age	integer	18 to 70 years	Customer age in completed years.
gender	string	Female, Male, Other	Customer demographic self-identification.
subscription_type	string	Basic, Standard, Premium	Current Netflix subscription tier.
watch_hours	float	0.10 to 110.40 hrs	Total cumulative monthly streaming hours.
last_login_days	integer	0 to 60 days	Number of days elapsed since the subscriber last logged in.
region	string	Africa, Asia, Europe, North America, Oceania, South America	Primary geographic operational market.
device	string	Desktop, Laptop, Mobile, TV, Tablet	Primary viewing hardware platform.

Attributes Description (Part 2)

monthly_fee	float	8.99, 13.99, 17.99 (USD)	Regular monthly recurring fee billed to account.
churned	integer	0 (Active), 1 (Churned)	Target variable: whether subscriber cancelled service.
payment_method	string	Credit Card, Crypto, Debit Card, Gift Card, PayPal	Payment method registered on the billing profile.
number_of_profiles	integer	1 to 5	Active viewer sub-profiles on account.
avg_watch_time_per_day	float	0.01 to 4.97 hrs	Calculated daily streaming engagement average.
favorite_genre	string	Action, Comedy, Drama, Horror, Romance, Sci-Fi, Thriller	Most frequently streamed content genre.



Entries Showcase

	customer_id	age	gender	subscription_type	watch_hours	last_login_days	region	device	monthly_fee
0	a9b75100-82a8-427a-a208-72f24052884a	51	Other	Basic	14.73	29	Africa	TV	8.99
1	49a5dfd9-7e69-4022-a6ad-0a1b9767fb5b	47	Other	Standard	0.70	19	Europe	Mobile	13.99
2	4d71f6ce-fca9-4ff7-8afa-197ac24de14b	27	Female	Standard	16.32	10	Asia	TV	13.99
3	d3c72c38-631b-4f9e-8a0e-de103cad1a7d	53	Other	Premium	4.51	12	Oceania	TV	17.99
4	4e265c34-103a-4dbb-9553-76c9aa47e946	56	Other	Standard	1.89	13	Africa	Mobile	13.99

churned	payment_method	number_of_profiles	avg_watch_time_per_day	favorite_genre
1	Gift Card	1	0.49	Action
1	Gift Card	5	0.03	Sci-Fi
0	Crypto	2	1.48	Drama
1	Crypto	2	0.35	Horror
1	Crypto	2	0.13	Action



Dataset dimensions

```
Dataset shape: (5000, 14)
Number of observations: 5000
Number of variables: 14
```

```
-----
```

Variable information:

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 5000 entries, 0 to 4999
```

```
Data columns (total 14 columns):
```

#	Column	Non-Null Count	Dtype
0	customer_id	5000 non-null	object
1	age	5000 non-null	int64
2	gender	5000 non-null	object
3	subscription_type	5000 non-null	object
4	watch_hours	5000 non-null	float64
5	last_login_days	5000 non-null	int64
6	region	5000 non-null	object
7	device	5000 non-null	object
8	monthly_fee	5000 non-null	float64
9	churned	5000 non-null	int64
10	payment_method	5000 non-null	object
11	number_of_profiles	5000 non-null	int64
12	avg_watch_time_per_day	5000 non-null	float64
13	favorite_genre	5000 non-null	object

```
dtypes: float64(3), int64(4), object(7)
```

A decorative graphic in the top-left corner consisting of two overlapping parallelograms: a blue one in the foreground and a light green one behind it. The background is a dark navy blue with faint, lighter blue diagonal stripes.

Data Preparation



Attributes pruning

	age	gender	subscription_type	watch_hours	last_login_days	region	device	churned	payment_method	number_of_profiles	favorite_genre
0	51	Other	Basic	14.73	29	Africa	TV	1	Gift Card	1	Action
1	47	Other	Standard	0.70	19	Europe	Mobile	1	Gift Card	5	Sci-Fi
2	27	Female	Standard	16.32	10	Asia	TV	0	Crypto	2	Drama
3	53	Other	Premium	4.51	12	Oceania	TV	1	Crypto	2	Horror
4	56	Other	Standard	1.89	13	Africa	Mobile	1	Crypto	2	Action

Dataset shape: (5000, 11)

Number of observations: 5000

Number of variables: 11

Checking Missing Values and Duplicates

```
print(df.duplicated(keep='last'))
print("\nTotal number of duplicates:", df.duplicated().sum())
```

```
0      False
1      False
2      False
3      False
4      False
```

```
...
4995   False
4996   False
4997   False
4998   False
4999   False
```

```
Length: 5000, dtype: bool
```

```
Total number of duplicates: 0
```

```
print("NULL VALUES:")
display(missing_values_null)
```

```
print("\n\nNAN VALUES:")
display(missing_values_nan)
```

NULL VALUES:

	Missing values	Missing percentage (%)
age	0	0.0
gender	0	0.0
subscription_type	0	0.0
watch_hours	0	0.0
last_login_days	0	0.0
region	0	0.0
device	0	0.0
churned	0	0.0
payment_method	0	0.0
number_of_profiles	0	0.0
favorite_genre	0	0.0

Text standardization

```
for col in df.columns:
    print(df[col].value_counts())
    print("-"*30)
```

```
44    96
10    94
...
38    68
19    68
17    65
15    62
14    60
```

Name: count, Length: 61, dtype: int64

```
-----
region
South America    873
Europe           867
North America    851
Asia             841
Africa           803
Oceania          765
```

Name: count, dtype: int64

```
-----
device
Tablet          1048
Laptop          1006
Mobile          1004
TV              993
Desktop         949
```

Name: count, dtype: int64

```
text_columns = df.select_dtypes(include=['object']).columns
for col in text_columns:
    df[col] = df[col].astype(str).str.strip().str.lower()
display(df.head())
```

	age	gender	subscription_type	watch_hours	last_login_days	region
0	51	other	basic	14.73	29	africa
1	47	other	standard	0.70	19	europa
2	27	female	standard	16.32	10	asia
3	53	other	premium	4.51	12	oceania
4	56	other	standard	1.89	13	africa

A decorative graphic on the left side of the slide. It consists of a blue parallelogram and a light green parallelogram, both tilted at an angle. The blue shape is in the foreground, and the green shape is partially behind it. They are set against a dark blue background with faint, lighter blue diagonal stripes.

Descriptive Statistics

Description of Discrete Variables

	Variable	Count	Mean	Median	Mode	Minimum	Q1 (25%)	Q2 (50%)	Q3 (75%)	Maximum
0	age	5000	43.847	44.0	63.00	18.00	30.000	44.0	58.00	70.0
1	watch_hours	5000	11.649	8.0	0.24	0.01	3.338	8.0	16.03	110.4
2	last_login_days	5000	30.090	30.0	37.00	0.00	15.000	30.0	45.00	60.0
3	number_of_profiles	5000	3.024	3.0	5.00	1.00	2.000	3.0	4.00	5.0

Range	Interquartile Range (IQR)	Variance	Standard Deviation	Skewness	Kurtosis
52.00	28.000	240.285	15.501	0.006	-1.244
110.39	12.692	144.352	12.015	2.259	7.797
60.00	30.000	307.514	17.536	-0.026	-1.190
4.00	2.000	2.005	1.416	-0.017	-1.303



Outliers Handling (*Detection*)

	Variable	Q1	Q3	IQR	Lower bound	Upper bound	Potential outliers	Percentage (%)
0	age	30.0000	58.00	28.0000	-54.00	142.0000	0	0.00
1	watch_hours	3.3375	16.03	12.6925	-34.74	54.1075	57	1.14
2	last_login_days	15.0000	45.00	30.0000	-75.00	135.0000	0	0.00
3	number_of_profiles	2.0000	4.00	2.0000	-4.00	10.0000	0	0.00

	Variable	z-score outliers	IQR outliers	Difference
0	age	0	0	0
1	watch_hours	102	57	45
2	last_login_days	0	0	0
3	number_of_profiles	0	0	0

Outliers Handling (*Detection*)



Outliers Handling (*Deletion*)

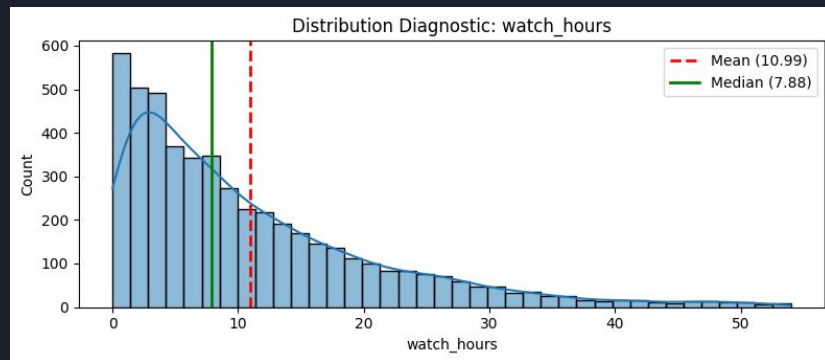
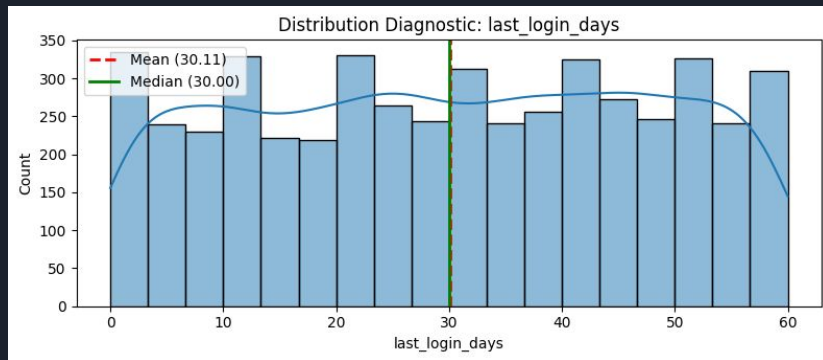
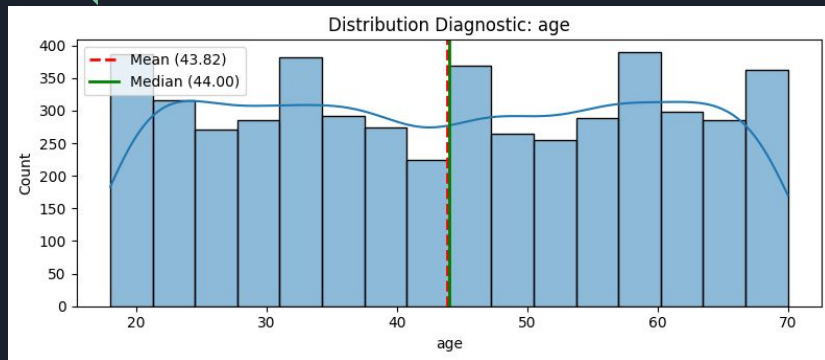
Capped upper limit for watch_hours: 54.98 hrs

watch_hours

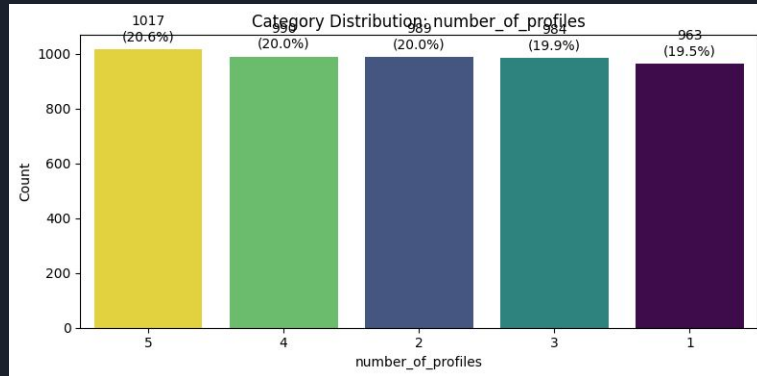
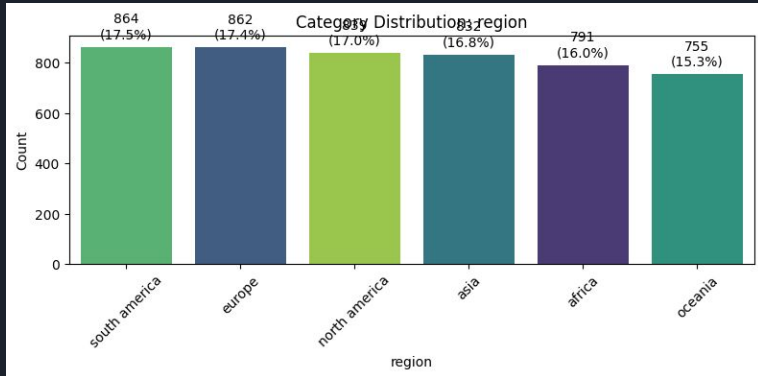
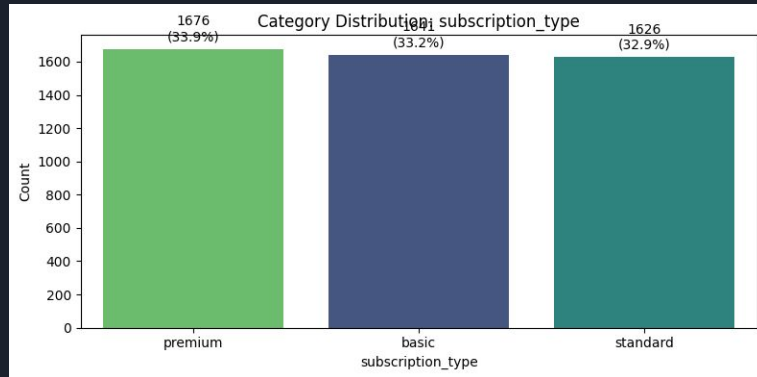
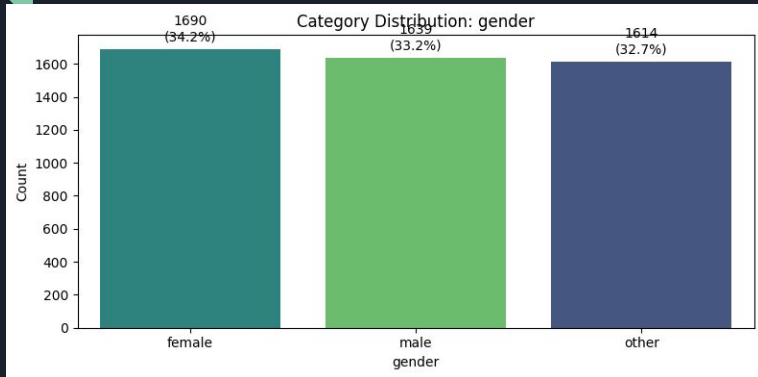


count	5000.000000
mean	11.488277
std	11.226075
min	0.010000
25%	3.337500
50%	8.000000
75%	16.030000
max	54.981500

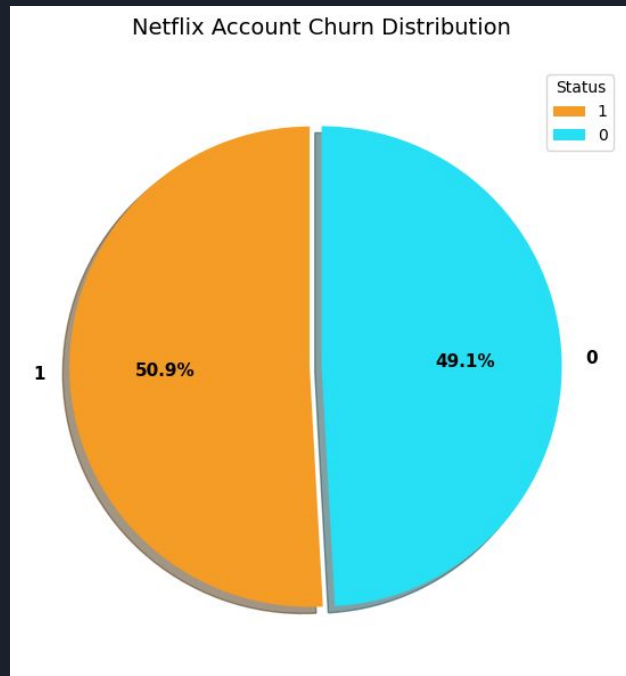
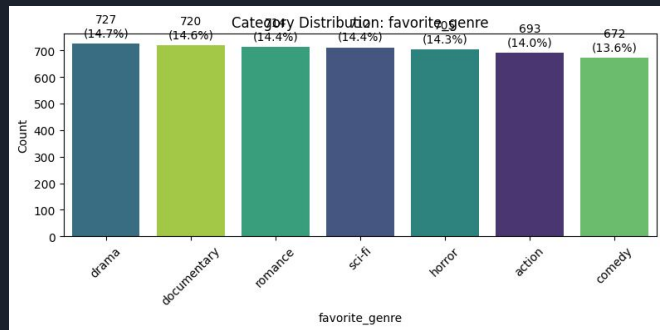
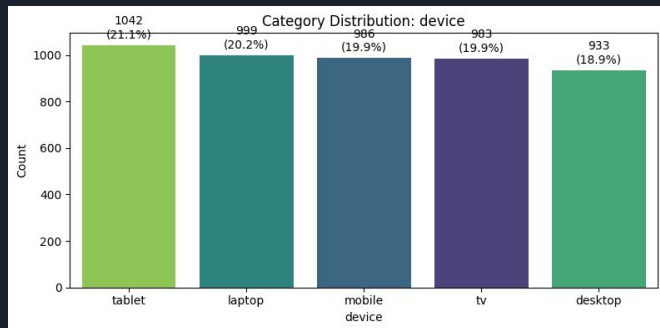
Visual Analysis of Attributes



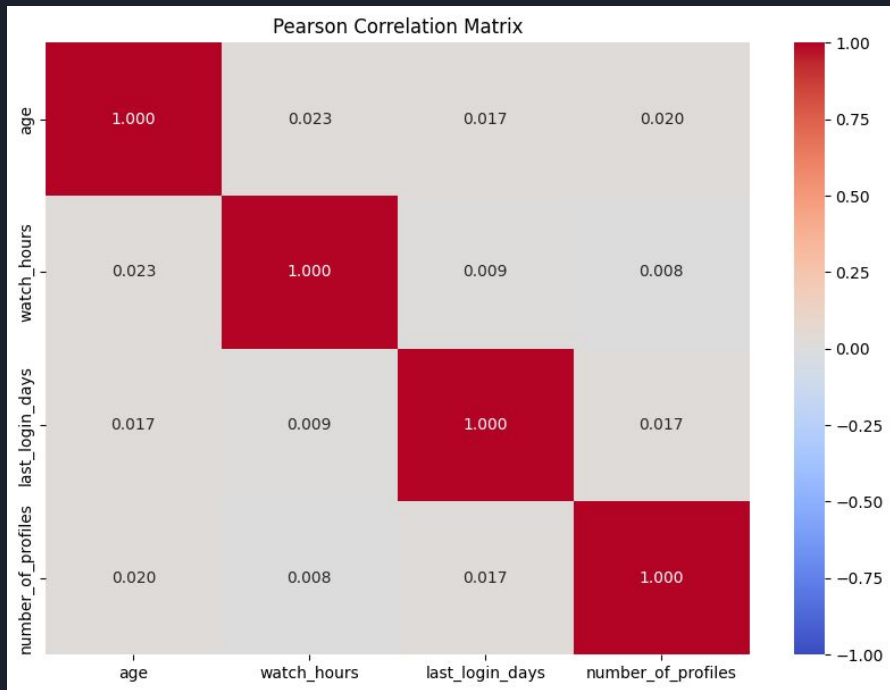
Visual Analysis of Attributes



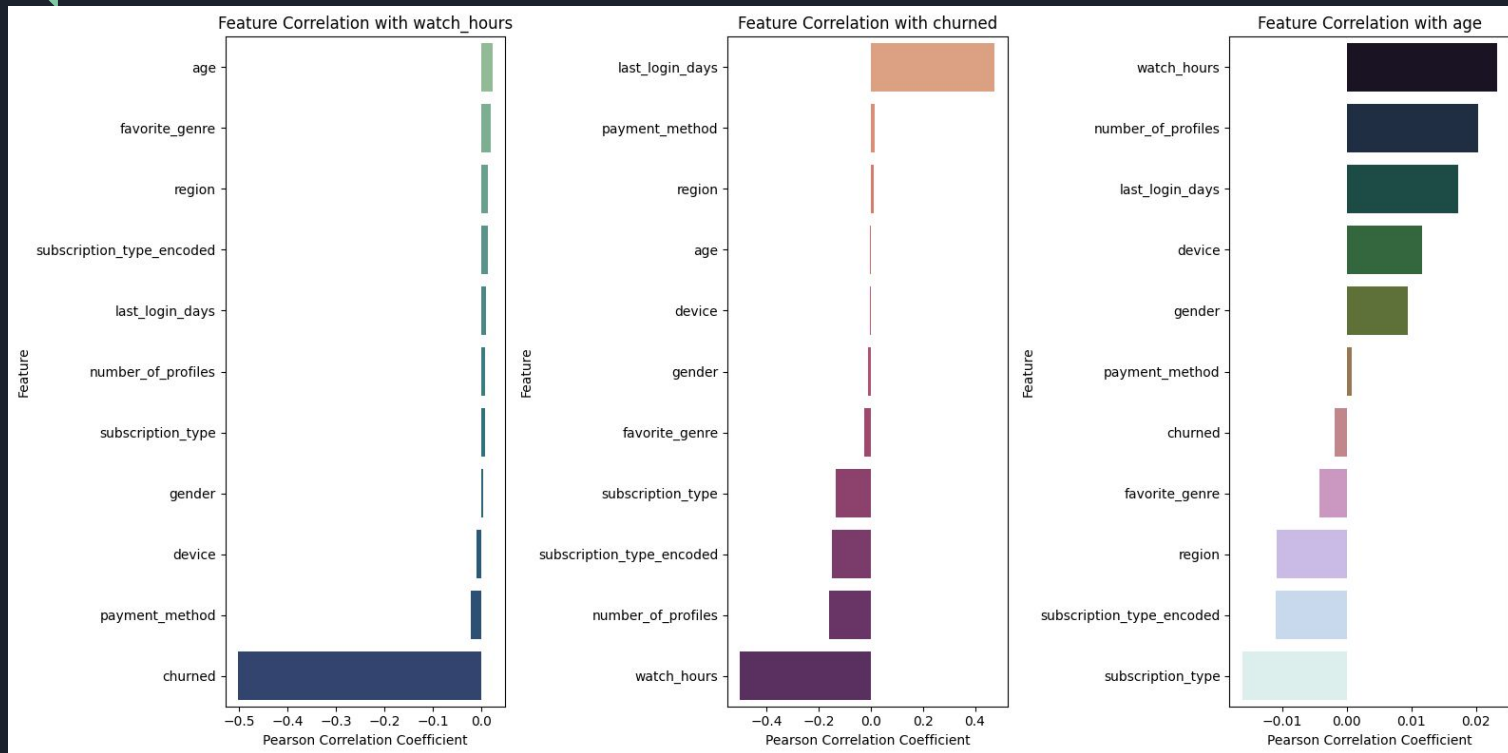
Visual Analysis of Attributes



Pearson Correlation for Integers



Pearson Correlation for Targets



A blue parallelogram and a light green parallelogram are positioned in the upper-left corner of the slide. The background is a dark navy blue with several diagonal bands of a slightly lighter shade of navy blue running from the bottom-left towards the top-right.

Hypothesis testing



One-Sample T-test for the population mean

One Sample T-Test

Is the population average monthly streaming time (watch_hours) equal to 11.0 hours? $\begin{cases} H_0 : \mu = 11.0 \\ H_1 : \mu \neq 11.0 \end{cases}$

```
=====
One sample t-test
-----
```

```
H0:  $\mu = 11.0$ 
```

```
H1:  $\mu \neq 11.0$ 
```

```
t = 3.0756
```

```
p-value = 0.002112
```

```
alpha = 0.05
```

```
Decision: Reject H0
```

```
Interpretation: There is statistically significant evidence against H0.
```

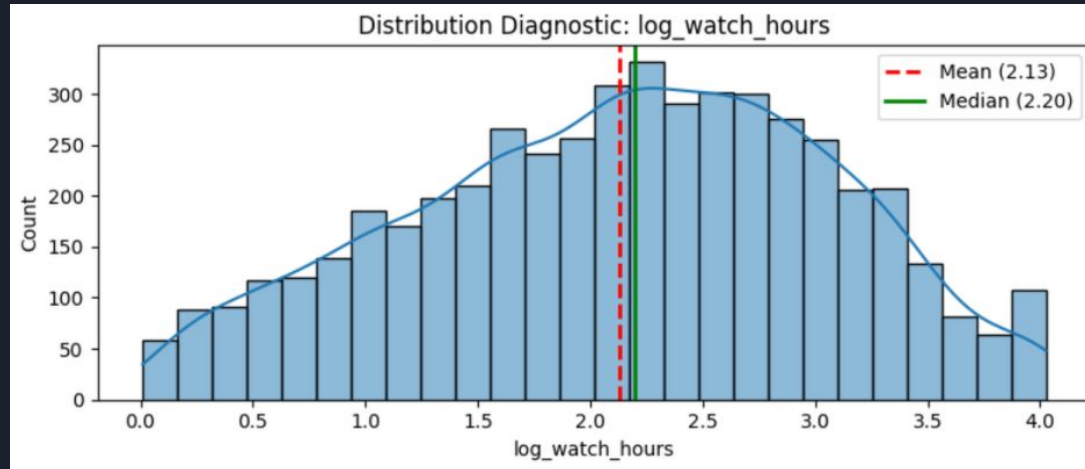
```
Sample mean = 11.49
```

```
Sample standard deviation = 11.23
```

```
Sample size = 5000
```

One-Sample T-test for the population mean

```
df["log_watch_hours"] = np.log1p(df["watch_hours"])  
display(df.head())
```





One-Sample T-test for the population mean

```
=====
Shapiro-Wilk test for normality
-----
H0: The population distribution is normal
H1: The population distribution is not normal
w = 0.9856
p-value = 0.000000
alpha = 0.05
Decision: Reject H0
Interpretation: There is statistically significant evidence against H0.
```

```
=====
One sample t-test
-----
H0:  $\mu = 2.0$ 
H1:  $\mu \neq 2.0$ 
t = 9.9570
p-value = 0.000000
alpha = 0.05
Decision: Reject H0
Interpretation: There is statistically significant evidence against H0.
Sample mean = 2.13
Sample standard deviation = 0.93
Sample size = 5000
```



One-Sample T-test for the population proportion

Hypothesis test for the population proportion

Is the population proportion of churn rate equal to **0.50** (or 50%)?

$$\begin{cases} H_0 : p = 0.50 \\ H_1 : p \neq 0.50 \end{cases}$$

```
=====
One sample population proportion test
-----
```

```
H0: p = 0.50
```

```
H1: p ≠ 0.50
```

```
z = 0.4243
```

```
p-value = 0.671368
```

```
alpha = 0.05
```

```
Decision: Do not reject H0
```

```
Interpretation: There is not sufficient statistical evidence against H0.
```

```
Observed proportion (churned = 1) = 0.5030
```

```
Total churned subscribers = 2515
```

```
Total subscribers (n) = 5000
```



Hypothesis testing for correlation

Hypothesis testing for correlation

Is there a statistically significant linear correlation between the **watch_hours** of customers and their **last_login_days** amount?

$$\begin{cases} H_0 : \rho = 0 \\ H_1 : \rho \neq 0 \end{cases}$$

Variables:

- `watch_hours`
- `last_login_days`

The Shapiro-Wilk test is performed for both variables before testing for correlation.

Hypothesis testing for correlation

```
... =====
Shapiro-Wilk test for normality: watch_hours
-----
H0: The population distribution is normal
H1: The population distribution is not normal
w = 0.8299
p-value = 0.000000
alpha = 0.05
Decision: Reject H0
Interpretation: There is statistically significant evidence against H0.
=====
Shapiro-Wilk test for normality: last_login_days
-----
H0: The population distribution is normal
H1: The population distribution is not normal
w = 0.9550
p-value = 0.000000
alpha = 0.05
Decision: Reject H0
Interpretation: There is statistically significant evidence against H0.
```



Hypothesis testing for correlation

```
... =====  
Hypothesis test for correlation  
-----  
H0:  $\rho = 0$   
H1:  $\rho \neq 0$   
 $r = 0.0033$   
 $p\text{-value} = 0.816948$   
 $\alpha = 0.05$   
Decision: Do not reject H0  
Interpretation: There is not sufficient statistical evidence against H0.  
 $n = 5000$ 
```



Equality of two population proportions

Equality of two population proportions

Are the population proportions of customers with inactive accounts buying **basic subscription** and **premium subscription** equal?

$$\begin{cases} H_0 : p_A = p_B \\ H_1 : p_A \neq p_B \end{cases}$$

Variables:

- `subscription_type = basic`
- `subscription_type = premium` for `churned`



Equality of two population proportions

```
=====
Equality of two population proportions: Churn Rate (Basic vs Premium)
-----
```

```
H0:  $p_A = p_B$ 
```

```
H1:  $p_A \neq p_B$ 
```

```
z = 10.5091
```

```
p-value = 0.000000
```

```
alpha = 0.05
```

```
Decision: Reject H0
```

```
Interpretation: There is statistically significant evidence against H0.
```

```
Basic tier churn rate = 0.6183 (n=1661)
```

```
Premium tier churn rate = 0.4371 (n=1693)
```



Chi-Square goodness-of-fit test

Chi-square (χ^2) goodness-of-fit test

Is the subscriber distribution across subscription_type evenly split across tiers (33.3% Basic, 33.3% Standard, 33.3% Premium)?

- Basic = 0.33 (33%)
- Standard = 0.33 (33%)
- Premium = 0.33 (33%)

$$\begin{cases} H_0 : p_1 = 0.33, p_2 = 0.33, p_3 = 0.33 \\ H_1 : \text{At least one } p_i \neq p_i^0 (i = 1, 2, \dots, n) \end{cases}$$

Variable: `subscription_type`

Chi-Square goodness-of-fit test

```
... =====
Chi-square goodness-of-fit test: Subscription Tiers
-----
H0: p_Basic = 0.333, p_Standard = 0.333, p_Premium = 0.333
H1: At least one tier proportion differs
 $\chi^2 = 0.6916$ 
p-value = 0.707654
alpha = 0.05
Decision: Do not reject H0
Interpretation: There is not sufficient statistical evidence against H0.
```

	Category	Observed	Expected	Observed Proportion	Expected Proportion
0	basic	1661	1666.666667	0.3322	0.333333
1	standard	1646	1666.666667	0.3292	0.333333
2	premium	1693	1666.666667	0.3386	0.333333





Chi-Square test of independence

Chi-square (χ^2) test of independence

Is **churn** status independent of having a specific **payment** option?

$\begin{cases} H_0 : \text{Payment method and Churn status are independent} \\ H_1 : \text{Payment method and Churn status are dependent} \end{cases}$

Variables:

- `payment_method`
- `churned`



Chi-Square test of independence

```
•• =====  
Chi-square test of independence: Payment Method vs Churn  
-----
```

```
H0: Payment Method and Churn are independent
```

```
H1: Payment Method and Churn are dependent
```

```
 $\chi^2 = 96.9069$ 
```

```
p-value = 0.000000
```

```
alpha = 0.05
```

```
Decision: Reject H0
```

```
Interpretation: There is statistically significant evidence against H0.
```

```
Observed frequencies:
```

Chi-Square test of independence

Observed frequencies:

churned	0	1
payment_method		
credit card	549	424
crypto	401	594
debit card	580	450
gift card	412	564
paypal	543	483

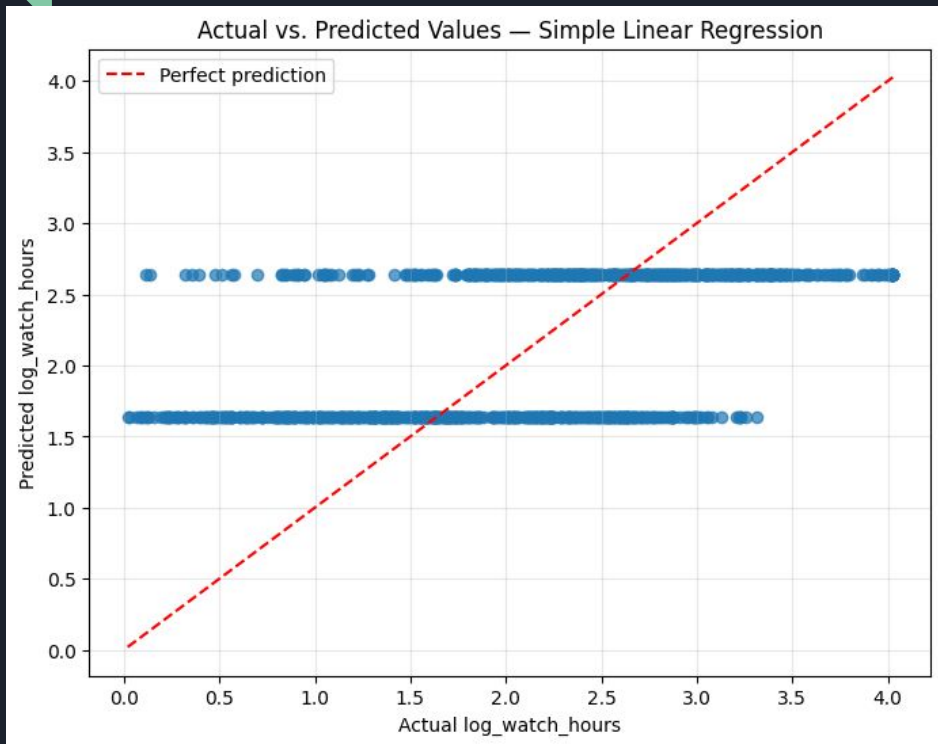
Expected frequencies:

churned	0	1
payment_method		
credit card	483.581	489.419
crypto	494.515	500.485
debit card	511.910	518.090
gift card	485.072	490.928
paypal	509.922	516.078

A decorative graphic on the left side of the slide consisting of two overlapping parallelograms. The front one is blue and the back one is a light green color. They are positioned diagonally, with the blue one in front of the green one.

Linear Regression

Single-Variable Regression



Feature: churned

	Metric	Value
--	--------	-------

0	MAE	0.637707
---	-----	----------

1	MSE	0.609614
---	-----	----------

2	RMSE	0.780778
---	------	----------

3	MAPE (%)	80.019671
---	----------	-----------

4	R-squared	0.274133
---	-----------	----------

Multi-Variable Regression



F-test for the overall regression model

F-statistic = 530.1287

p-value = 0.000000

alpha = 0.05

Reject H0: The overall regression model is statistically significant.

Multiple Linear Regression Model Coefficients:

Intercept = 2.714176

Age coefficient = -0.083491

Last Login Days coefficient = -1.381626

Equation: $\text{watch_hours} = 2.7142 + (-0.083491 \times \text{age}) + (-1.381626 \times \text{last_login_days})$

OLS Regression Results

Dep. Variable:	log_watch_hours	R-squared:	0.399
Model:	OLS	Adj. R-squared:	0.398
Method:	Least Squares	F-statistic:	530.1
Date:	Thu, 08 Oct 2026	Prob (F-statistic):	0.00
Time:	17:40:40	Log-Likelihood:	-4387.1
No. Observations:	4000	AIC:	8786.
Df Residuals:	3994	BIC:	8824.
Df Model:	5		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	2.7142	0.055	49.305	0.000	2.606	2.822
number_of_profiles	-0.0835	0.008	-10.117	0.000	-0.100	-0.067
churned	-1.3816	0.027	-51.463	0.000	-1.434	-1.329
age	0.0007	0.001	0.903	0.367	-0.001	0.002
last_login_days	0.0188	0.001	25.065	0.000	0.017	0.020
subscription_type_encoded	-0.1124	0.014	-7.960	0.000	-0.140	-0.085

Omnibus:

Prob(Omnibus):

Skew:

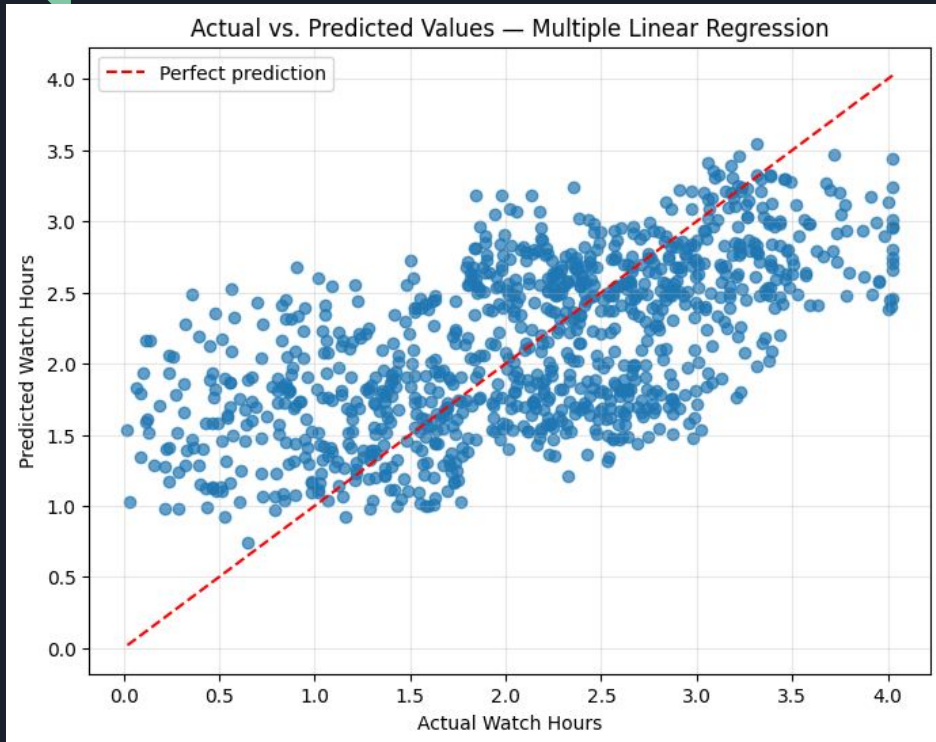
Kurtosis:

Cond. No.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Multi-Variable Regression

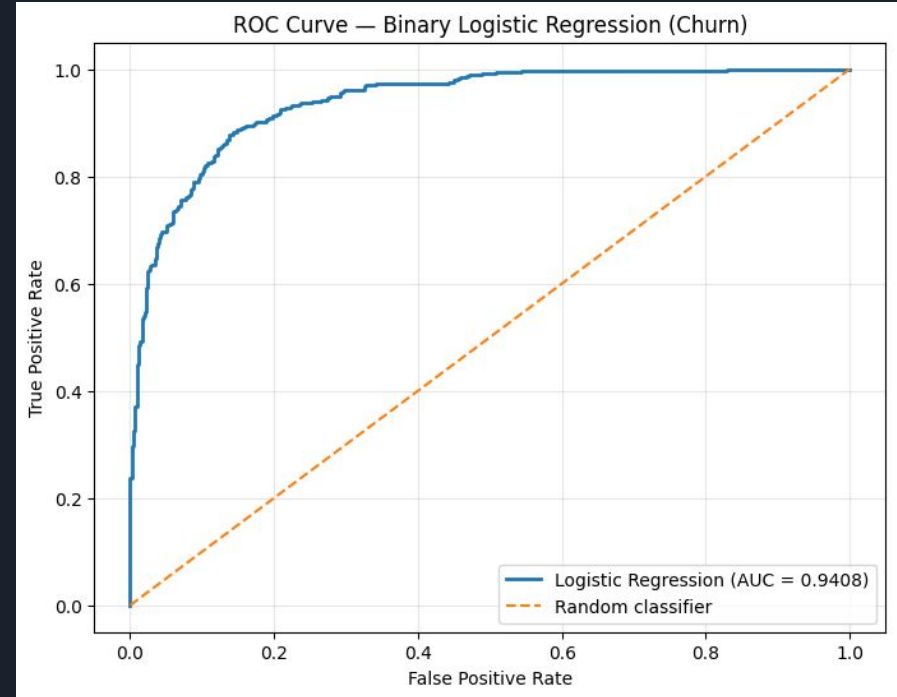
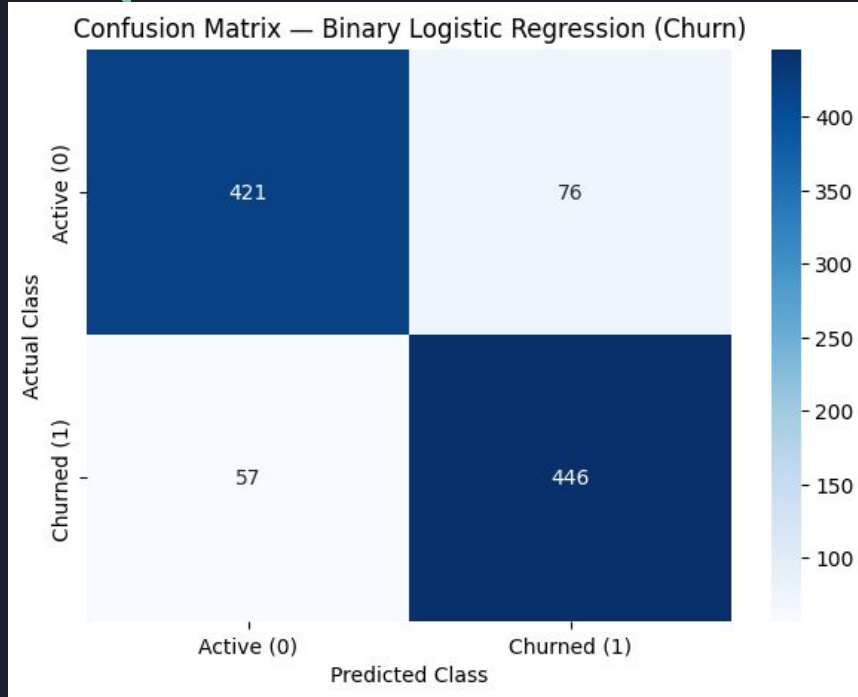


	Metric	Value
0	MAE	0.579880
1	MSE	0.516994
2	RMSE	0.719023
3	MAPE (%)	71.784946
4	R-squared	0.384414

A decorative graphic on the left side of the slide consisting of two overlapping parallelograms. The front one is blue and the back one is light green. They are positioned diagonally, with the blue one partially covering the green one.

Logistic Regression

Confusion Matrix & AUC





Statistics

```
... Classification report:
              precision    recall  f1-score   support

   Active      0.88        0.85        0.86        497
   Churned     0.85        0.89        0.87        503

 accuracy              0.87        1000
 macro avg              0.87        1000
 weighted avg           0.87        1000
```



New User Prediction

Input subscriber details:

	watch_hours	last_login_days	number_of_profiles
--	-------------	-----------------	--------------------

0	30.0	100	5
---	------	-----	---

Probability of remaining Active: 0.1402 (14.02%)

Probability of Churning: 0.8598 (85.98%)

Input subscriber details:

	watch_hours	last_login_days	number_of_profiles
--	-------------	-----------------	--------------------

0	5.0	10	3
---	-----	----	---

Probability of remaining Active: 0.7387 (73.87%)

Probability of Churning: 0.2613 (26.13%)

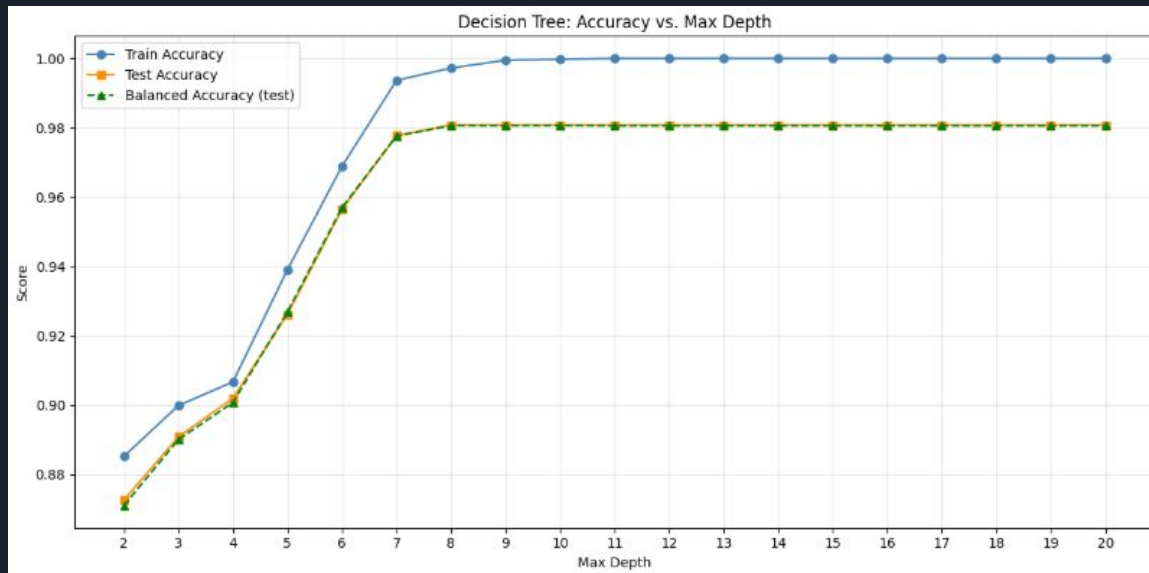
Predicted Subscriber State: Active

A decorative graphic in the top-left corner consisting of two overlapping parallelograms. The front one is blue and the back one is a light mint green. They are both tilted at a 45-degree angle.

Decision Trees

Decision Trees - Hyperparameter Tuning

	Max Depth	Train Accuracy	Test Accuracy	Balanced Accuracy	Macro F1
0	2	0.8852	0.8726	0.8710	0.8711
1	3	0.8998	0.8908	0.8900	0.8904
2	4	0.9067	0.9019	0.9007	0.9012
3	5	0.9390	0.9262	0.9269	0.9261
4	6	0.9689	0.9565	0.9570	0.9565
5	7	0.9937	0.9778	0.9776	0.9777
6	8	0.9972	0.9808	0.9806	0.9808
7	9	0.9995	0.9808	0.9806	0.9808
8	10	0.9997	0.9808	0.9807	0.9808
9	11	1.0000	0.9808	0.9806	0.9808
10	12	1.0000	0.9808	0.9806	0.9808
11	13	1.0000	0.9808	0.9806	0.9808
12	14	1.0000	0.9808	0.9806	0.9808
13	15	1.0000	0.9808	0.9806	0.9808
14	16	1.0000	0.9808	0.9806	0.9808
15	17	1.0000	0.9808	0.9806	0.9808
16	18	1.0000	0.9808	0.9806	0.9808
17	19	1.0000	0.9808	0.9806	0.9808
18	20	1.0000	0.9808	0.9806	0.9808



Decision Trees - Cross-Validation & Rules

=== Complex Decision Tree (max_depth=10, min_samples_leaf=5) ===
=== Stratified 10-Fold Cross-Validation Results ===

	Accuracy	Balanced Accuracy	Precision	Recall	F1 Score	G-Mean
Fold 1	0.9879	0.9877	0.9881	0.9877	0.9879	0.9877
Fold 2	0.9717	0.9716	0.9720	0.9716	0.9717	0.9716
Fold 3	0.9778	0.9780	0.9778	0.9780	0.9778	0.9780
Fold 4	0.9838	0.9836	0.9841	0.9836	0.9838	0.9836
Fold 5	0.9858	0.9860	0.9858	0.9860	0.9858	0.9860
Fold 6	0.9838	0.9838	0.9838	0.9838	0.9838	0.9838
Fold 7	0.9757	0.9753	0.9772	0.9753	0.9757	0.9753
Fold 8	0.9737	0.9738	0.9737	0.9738	0.9737	0.9738
Fold 9	0.9757	0.9758	0.9757	0.9758	0.9757	0.9758
Fold 10	0.9636	0.9639	0.9643	0.9639	0.9636	0.9639
Average	0.9779	0.9780	0.9782	0.9780	0.9779	0.9780

=== Rule-Based Decision Tree (max_depth=3) ===
=== Stratified 10-Fold Cross-Validation Results ===

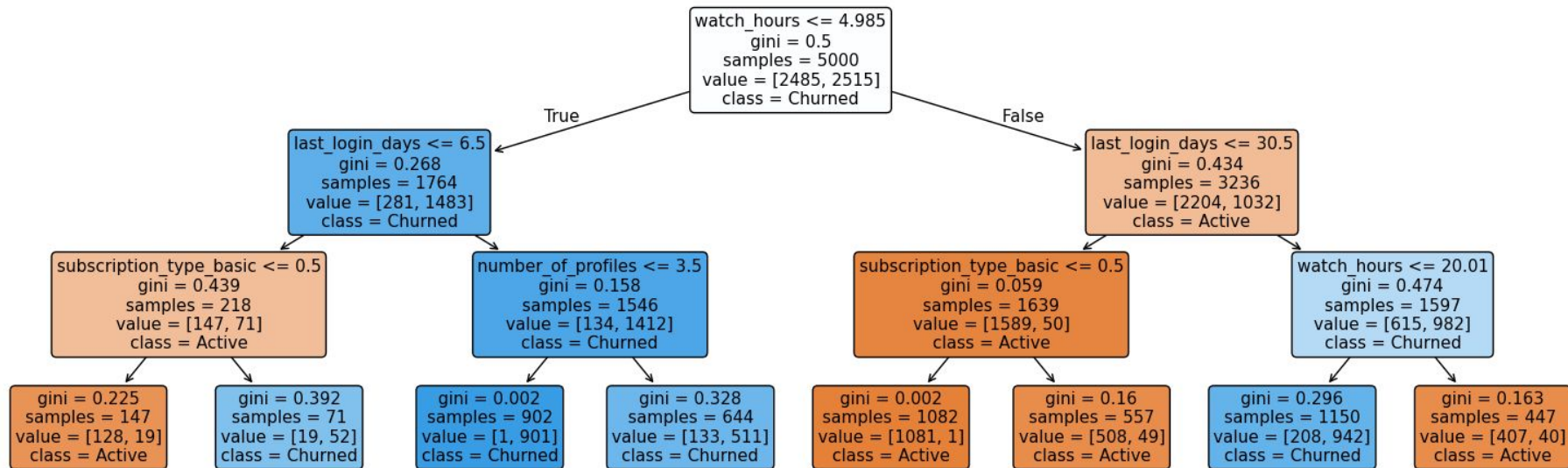
	Accuracy	Balanced Accuracy	Precision	Recall	F1 Score	G-Mean
Fold 1	0.9010	0.8998	0.9090	0.8998	0.9003	0.8998
Fold 2	0.8949	0.8943	0.8969	0.8943	0.8947	0.8943
Fold 3	0.9192	0.9186	0.9213	0.9186	0.9190	0.9186
Fold 4	0.9271	0.9261	0.9319	0.9261	0.9268	0.9261
Fold 5	0.9231	0.9220	0.9286	0.9220	0.9227	0.9220
Fold 6	0.8988	0.8981	0.9018	0.8981	0.8985	0.8981
Fold 7	0.8947	0.8937	0.9021	0.8937	0.8940	0.8937
Fold 8	0.9190	0.9185	0.9211	0.9185	0.9188	0.9185
Fold 9	0.8785	0.8780	0.8804	0.8780	0.8783	0.8780
Fold 10	0.8543	0.8535	0.8574	0.8535	0.8537	0.8535
Average	0.9011	0.9002	0.9050	0.9002	0.9007	0.9002

Extracted Decision Rules (max_depth=3)

```
--- watch_hours <= 4.98
|   --- last_login_days <= 6.50
|   |   --- subscription_type_basic <= 0.50
|   |   |   --- class: 0
|   |   --- subscription_type_basic > 0.50
|   |   |   --- class: 1
|   --- last_login_days > 6.50
|   |   --- number_of_profiles <= 3.50
|   |   |   --- class: 1
|   |   --- number_of_profiles > 3.50
|   |   |   --- class: 1
|   --- watch_hours > 4.98
|   |   --- last_login_days <= 30.50
|   |   |   --- subscription_type_basic <= 0.50
|   |   |   |   --- class: 0
|   |   |   --- subscription_type_basic > 0.50
|   |   |   |   --- class: 0
|   |   --- last_login_days > 30.50
|   |   |   --- watch_hours <= 20.01
|   |   |   |   --- class: 1
|   |   |   --- watch_hours > 20.01
|   |   |   |   --- class: 0
```

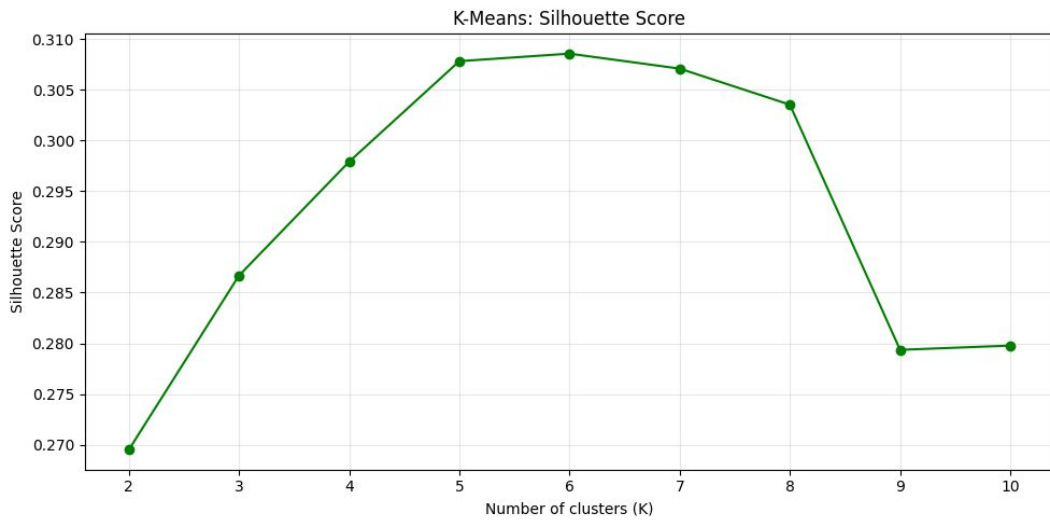
Decision Tree

Decision Tree — Rule-Based Classifier (max_depth = 3)

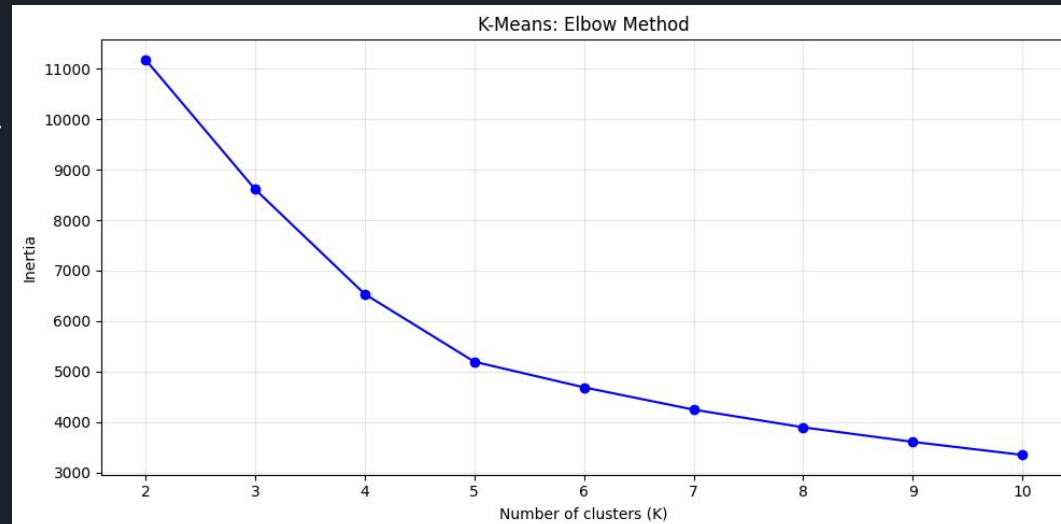


A decorative graphic on the left side of the slide consisting of two overlapping parallelograms. The front one is blue and the back one is a light greenish-blue. They are positioned diagonally, with the blue one partially covering the green one.

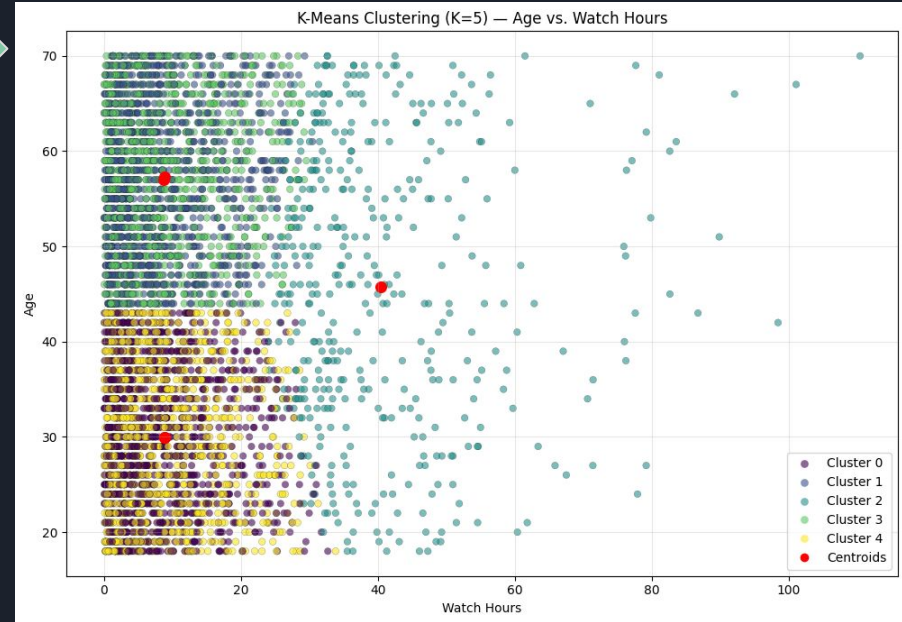
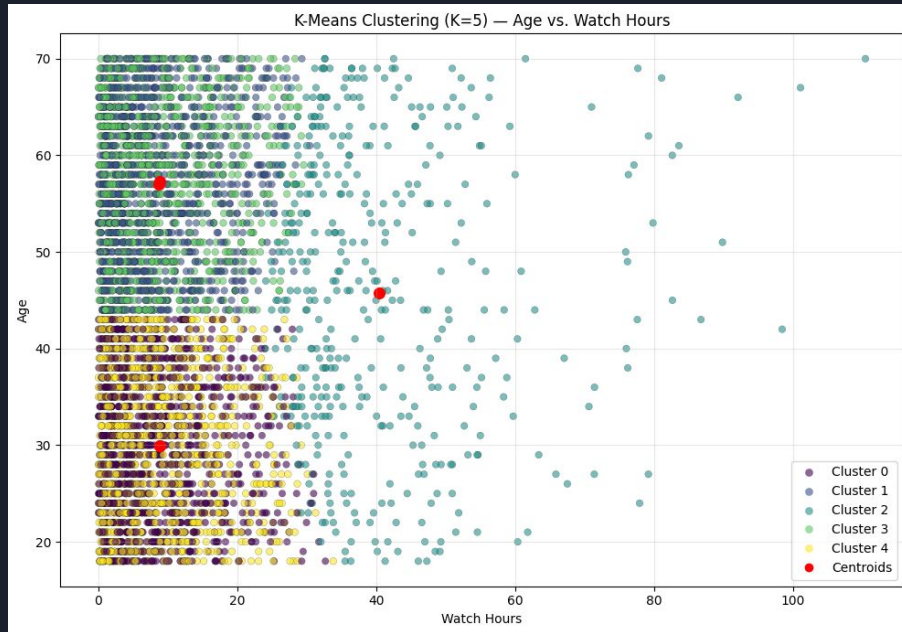
Clusters: K-Means & K-Medoid



Customer Segmentation - Choosing K



K-Means vs K-Medoids



Silhouette scores comparison:

K-means: 0.308

K-medoids: 0.302



References

Project's Github Link - https://github.com/michalmatysik6/Data_Science---Winter-school

Dataset "Netflix Customer Churn & Engagement Analytics" on Kaggle -

<https://www.kaggle.com/datasets/zeyadmohamed26/netflix-customer-churn-and-engagement-analytics/data> (Last Reviewed : October 8th 2026)